MILITARY STANDARD

RAILCAR LOADING OF HAZARDOUS MATERIALS

MIL-STD-1325-37
(NAVY)

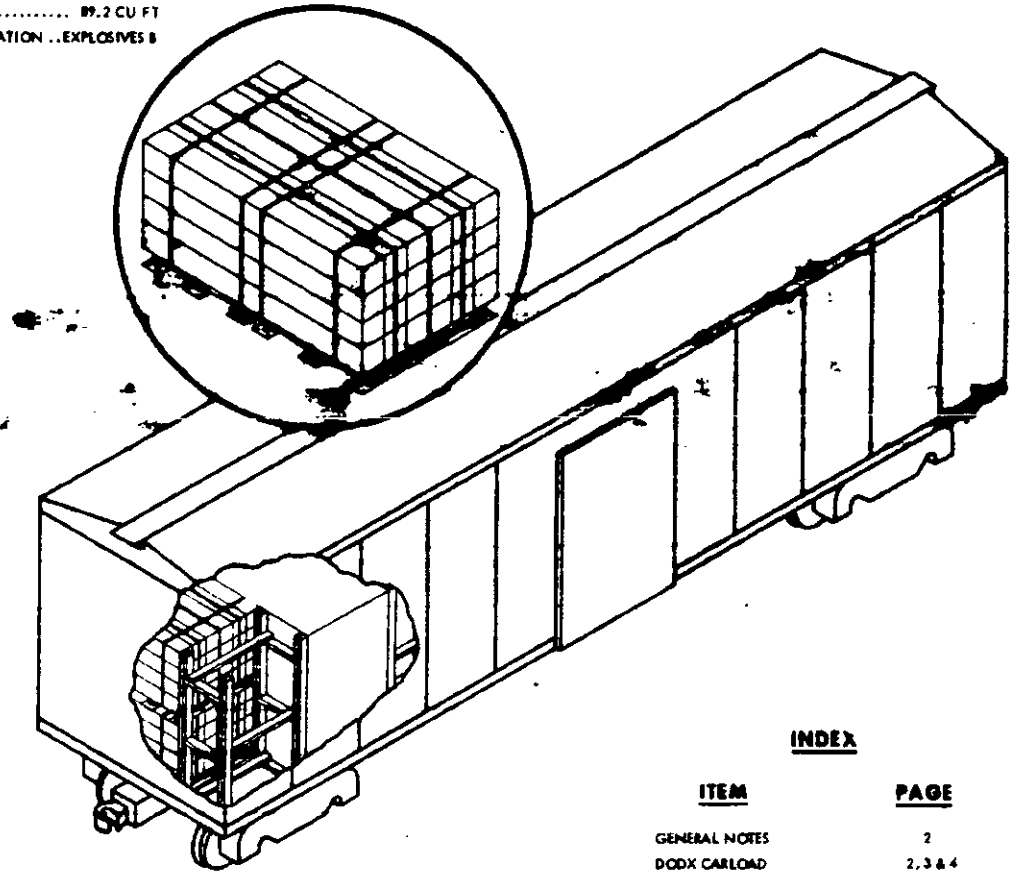
20 MAY 1974

SUPERSEDING WR-52/37 A
10 FEBRUARY 1970

ROCKET MOTOR MK 16 & MODS
OR MK 71 MOD 0 (ZUNI)
IN CONTAINER MK 38 MOD 0
FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/28
WEIGHT ..... SEE TABLE I
DIMENSIONS ... 81 1/8 L x 49 1/2 W x 38 3/8 H
CUBE ..... 89.2 CU FT
DOT HAZARD CLASSIFICATION ..EXPLOSIVES B



NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINER LOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY).

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FSC 8140

AUTHORIZED AND RELEASED GENERAL USE.

M. S. Gray NWHL 5/24/74
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J. E. Kelly 5/25/74
SIGNATURE (AIR) (SYSCOM) BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

C. J. Grossmuck 3/6/74
SIGNATURE SUPERVISOR, MILITARY & INTERMODEL SERVICES DATE

ORIGINATOR Charles MacBride 4/10/74
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

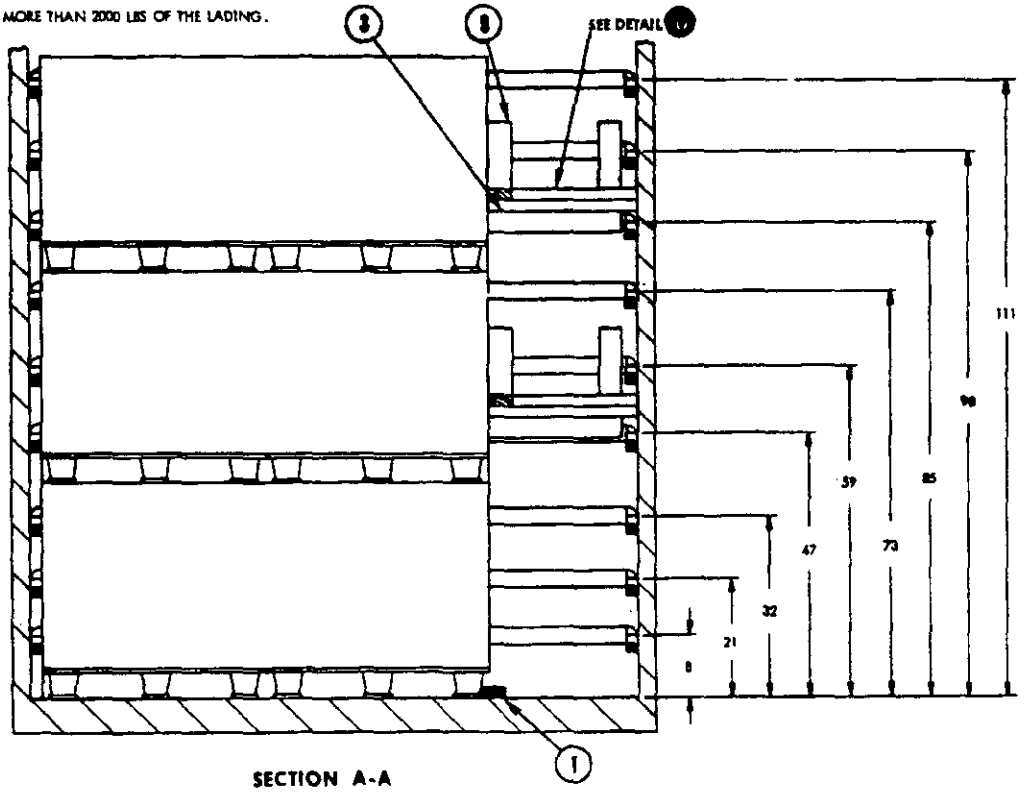
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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT WR-52 "CARLOADING OF WEAPONS AND MAJOR WEAPON SYSTEM COMPONENTS."
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 28000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), COMMERCIAL BOXCARS WITH 40 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH.
4. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
5. THE LOAD CONSISTS OF 5" ZUNI ROCKET MOTORS, MK 16 MOD 1, IN MK 38 CONTAINERS, PALLETIZED IN ACCORDANCE WITH WR-54/78.
6. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK. FOR THE ALTERNATE METHOD THE LOADS ARE HANDLED FROM THE END BY A FORK TRUCK WITH APPROVED FORK TINE EXTENSIONS OR A HAND PALLET TRUCK WHICH HAS THIS CAPABILITY.
7. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH WR-52.
8. APPLICABLE MATERIAL SPECIFICATIONS:
   - DUNNAGE LUMBER - FED SPEC MM-L-751
   - NAILS - FED SPEC FF-N-105
   - STRAPPING - FED SPEC QQ-S-781, TY I, CL A.
9. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF THE CAR AS PRESCRIBED IN OP 2165 (VOL 1).

50 FT 6 IN. BOXCAR, DODX

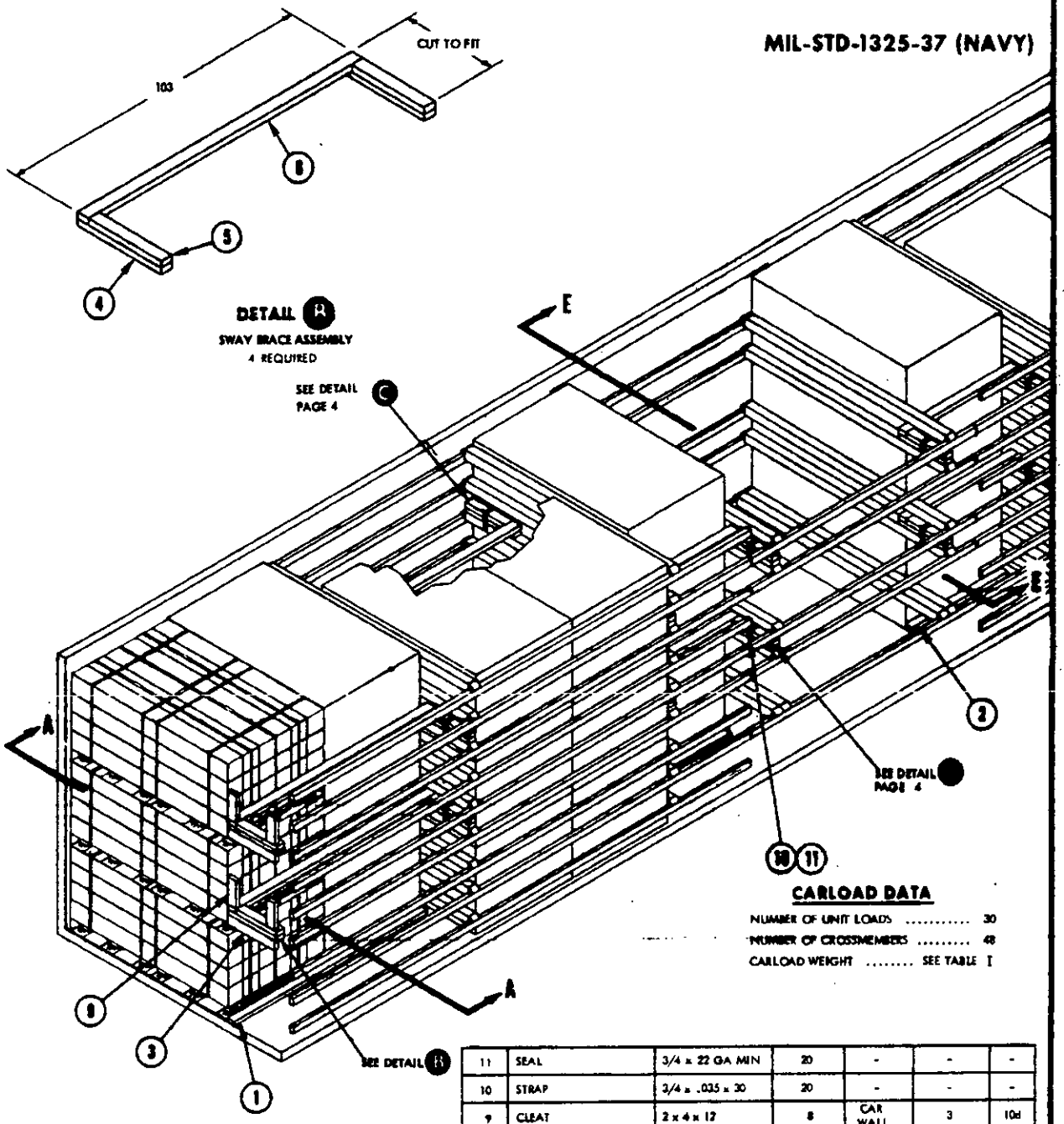
1. THE CARLOAD CONSISTS OF 30 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH (SEE WR-52). EACH CROSS MEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2000 LBS OF THE LADING.



SECTION A-A

SHOWING LOCATION OF WALLMEMBERS USED FOR CROSSMEMBERS

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CARLOAD DATA

NUMBER OF UNIT LOADS 30
NUMBER OF CROSSMEMBERS 48
CARLOAD WEIGHT ..... SEE TABLE I

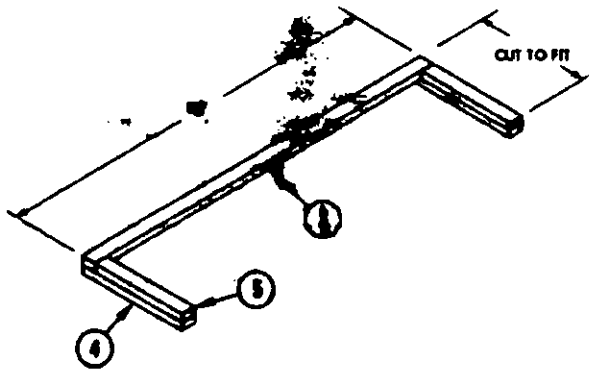
TABLE I

ROCKET MOTOR	APPROXIMATE WEIGHT IN LBS	
	UNIT LOAD	CARLOAD
MK 16 & MODS	2741	82630
MK 71 MOD 0	2813	84790
APPROXIMATE WEIGHT OF DUNNAGE (400 LBS) INCLUDED IN CARLOAD WEIGHT		

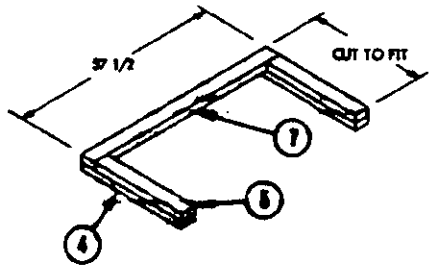
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS: NUMBER	NAILS: SIZE
11	SEAL	3/4 x 22 GA MIN	20	-	-	-
10	STRAP	3/4 x .035 x 30	20	-	-	-
9	CLEAT	2 x 4 x 12	8	CAR WALL	3	10d
8	SWAY BRACE STRINGER	2 x 4 x 57 1/2	4	4	3 PER JOINT	10d
7	SWAY BRACE STRINGER	2 x 4 x 107	4	4	3 PER JOINT	10d
6	SWAY BRACE STRINGER	2 x 4 x 103	4	4	3 PER JOINT	10d
5	BACK-UP CLEAT	2 x 4 x CUT TO FIT	24	4	4	10d
4	SWAY BRACE STRUT	2 x 4 x CUT TO FIT	24	SEE 5	-	-
3	CLEAT	2 x 4 x 24	4	CAR WALL	4	10d
2	SLEEPER	2 x 4 x 48	2	CAR FLOOR	2 PER FOOT	16d
1	SLEEPER	2 x 4 x 96	4	CAR FLOOR	2 PER FOOT	16d

LIST OF MATERIALS AND NAILING DATA

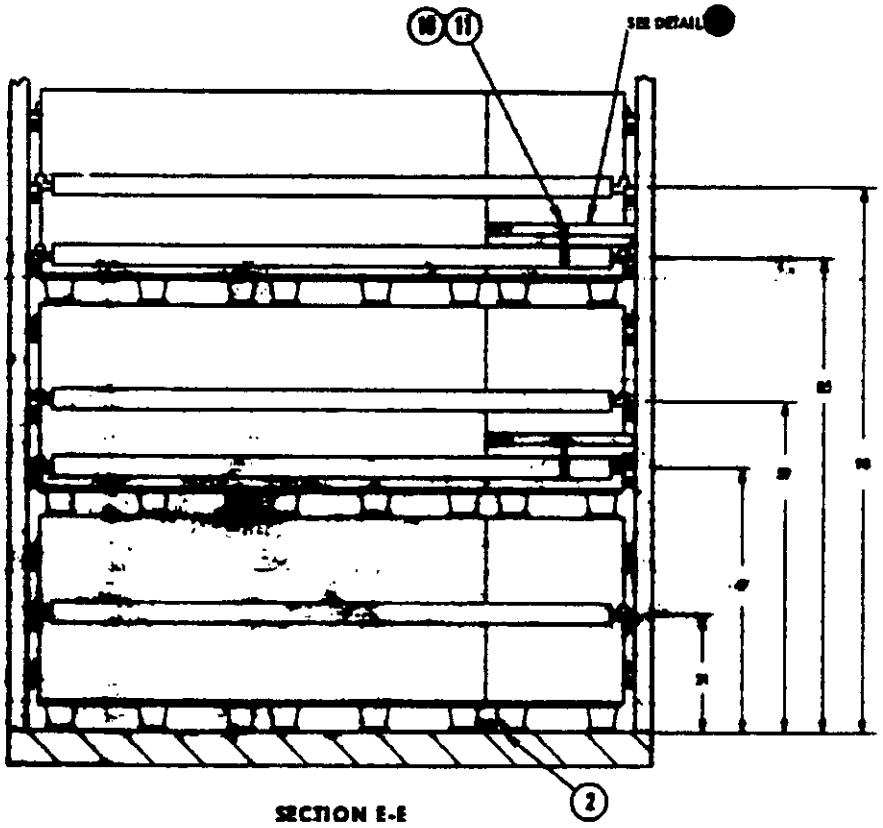
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DETAIL C
SWAY BRACE ASSEMBLY
4 REQUIRED



DETAIL D
SWAY BRACE ASSEMBLY
4 REQUIRED



SECTION E-E
SHOWING LOCATION OF DOORWAY MEMBERS

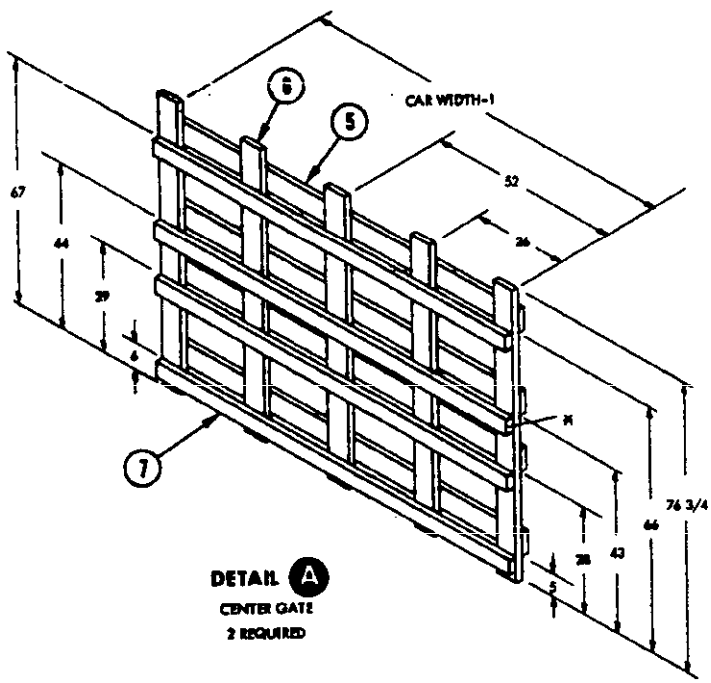
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40 FT 6 IN. BOXCAR, COMMERCIAL

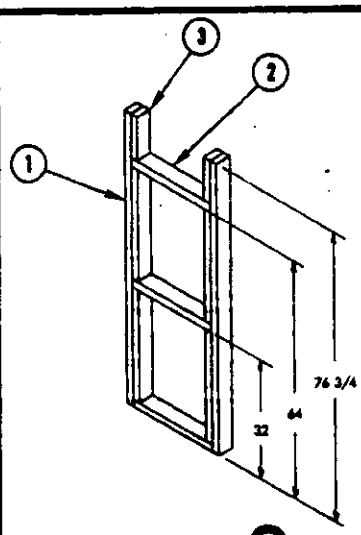
1. THE CARLOAD CONSISTS OF 16 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH DETAILS PAGES 8, 9 AND 10 OF WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE, PROVIDED DOORWAY PROTECTION WHEN REQUIRED IS INSTALLED IN ACCORDANCE WITH WR-52.
4. IF CAR HAS STEEL SIDEWALLS DELETE PIECE NUMBER 11 & 12. TO HOLD GATE ASSEMBLY DOWN NAIL A DOUBLED 2 x 4 x 5 FT HORIZONTAL TO THE GATE VERTICALS ON THE SIDE FACING THE LADING. POSITION HORIZONTALS WITH THE BOTTOM EDGE 39 INCHES FROM THE FLOOR (FIELD CHECK THIS DIMENSION) AND SO THAT THE HORIZONTAL IS TRAPPED UNDER THE UPPER UNIT LOAD. NAIL EACH JOINT WITH THREE 10d NAILS PER JOINT.



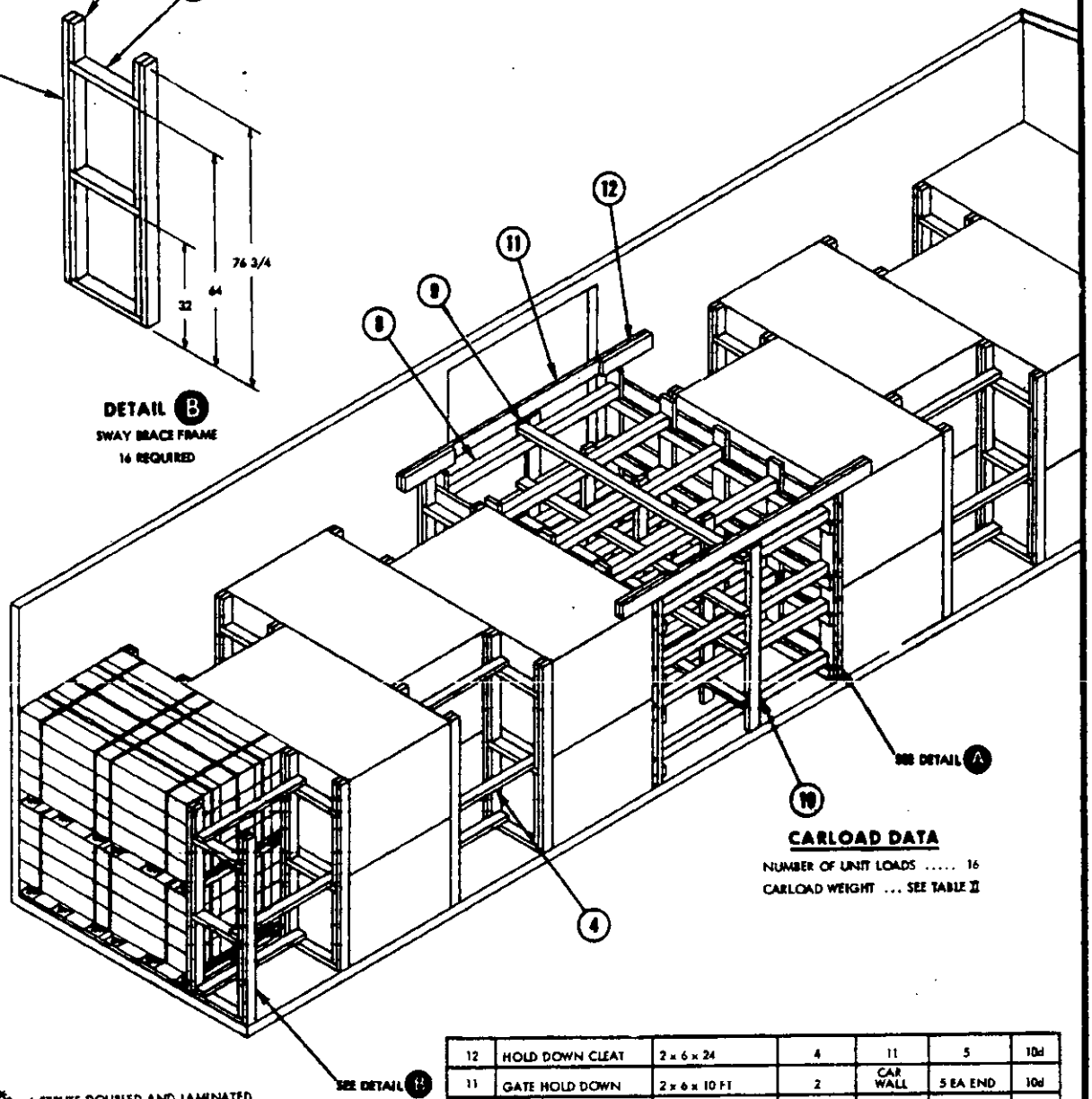
DETAIL A
CENTER GATE
2 REQUIRED

*POSITION AFTER STRUTS HAVE BEEN INSTALLED ON LOWER STRUT CLEAT. (REQUIRED ONE GATE ONLY)

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DETAIL B
SWAY BRACE FRAME
16 REQUIRED



CARLOAD DATA

NUMBER OF UNIT LOADS 16
CARLOAD WEIGHT ... SEE TABLE II

*2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

TABLE II

ROCKET MOTOR	APPROXIMATE WEIGHT IN LBS	
	UNIT LOAD	CARLOAD
MK 16 & MODS	2741	45966
MK 71 MOD 0	2813	47118
APPROXIMATE WEIGHT OF DUNNAGE (7110 LBS) INCLUDED IN CARLOAD WEIGHT		

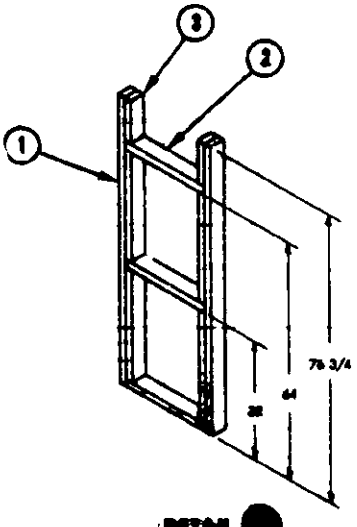
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE
12	HOLD DOWN CLEAT	2 x 6 x 24	4	11	5	10d
11	GATE HOLD DOWN	2 x 6 x 10 FT	2	CAR WALL	5 EA END	10d
10	VERTICAL TIE BAR	2 x 4 x 76	5	8	3 PER JOINT	10d
9	HORIZONTAL TIE BAR	2 x 4 x CAR WIDTH-1	4	8	3 PER JOINT	10d
8	STRUT	4 x 4 x WEDGE FIT *	20	6	2 PER JOINT	16d
7	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	6	3 PER JOINT	10d
6	CENTER GATE VERTICAL	2 x 6 x 76 3/4	10	SEE 5	-	-
5	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	6	3 PER JOINT	10d
4	SPREADER	2 x 4 x 49 1/2	24	2	2 PER JOINT	10d
3	SWAY BRACE CLEAT	2 x 4 x CUT TO FIT	96	1	3	10d
2	SWAY BRACE STRUT	2 x 4 x CUT TO FIT	48	SEE 3	-	-
1	SWAY BRACE VERTICAL	2 x 4 x 76 3/4	32	2	2 PER JOINT	16d

LIST OF MATERIALS AND NAILING DATA

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50 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 24 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH DETAILS PAGES 8, 9 AND 10 OF WR-S2/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING THE LENGTH OF PIECE 12 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 12 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR, AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. THE STACKS THAT ARE IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND A TOP SWAY BRACE CONSISTING OF A STRAPPING BOARD AND CROSS BRACE (SEE DETAIL F) MUST BE INSTALLED BETWEEN THE ROWS OF THE DOORWAY STACKS. STRAP IS SECURED TO STRAPPING BOARD WITH TWO 1 1/4" STAPLES. DOORWAY PROTECTION, PIECES 14 AND 15 IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.
5. IF CAR HAS STEEL SIDEWALLS DELETE PIECE NUMBERS 12 & 13. TO HOLD GATE ASSEMBLY DOWN NAIL A DOUBLED 2 x 4 x 5 FT. HORIZONTAL TO THE GATE VERTICALS ON THE SIDE FACING THE LADING. POSITION HORIZONTALS WITH THE BOTTOM EDGE 39 INCHES FROM THE FLOOR (FIELD CHECK THIS DIMENSION) AND SO THAT THE HORIZONTAL IS TRAPPED UNDER THE UPPER UNIT LOAD. NAIL EACH JOINT WITH THREE 10d NAILS PER JOINT.



DETAIL
SWAY BRACE FRAME
24 REQUIRED

*CLINCHED

**2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

TABLE III

ROCKET MOTOR	APPROXIMATE WEIGHT IN LBS	
	UNIT LOAD	CARLOAD
MK 16 & MODS	2741	68076
MK 71 MOD 0	2813	69804
APPROXIMATE WEIGHT OF DUNNAGE (1085 LBS) INCLUDED IN CARLOAD WEIGHT		

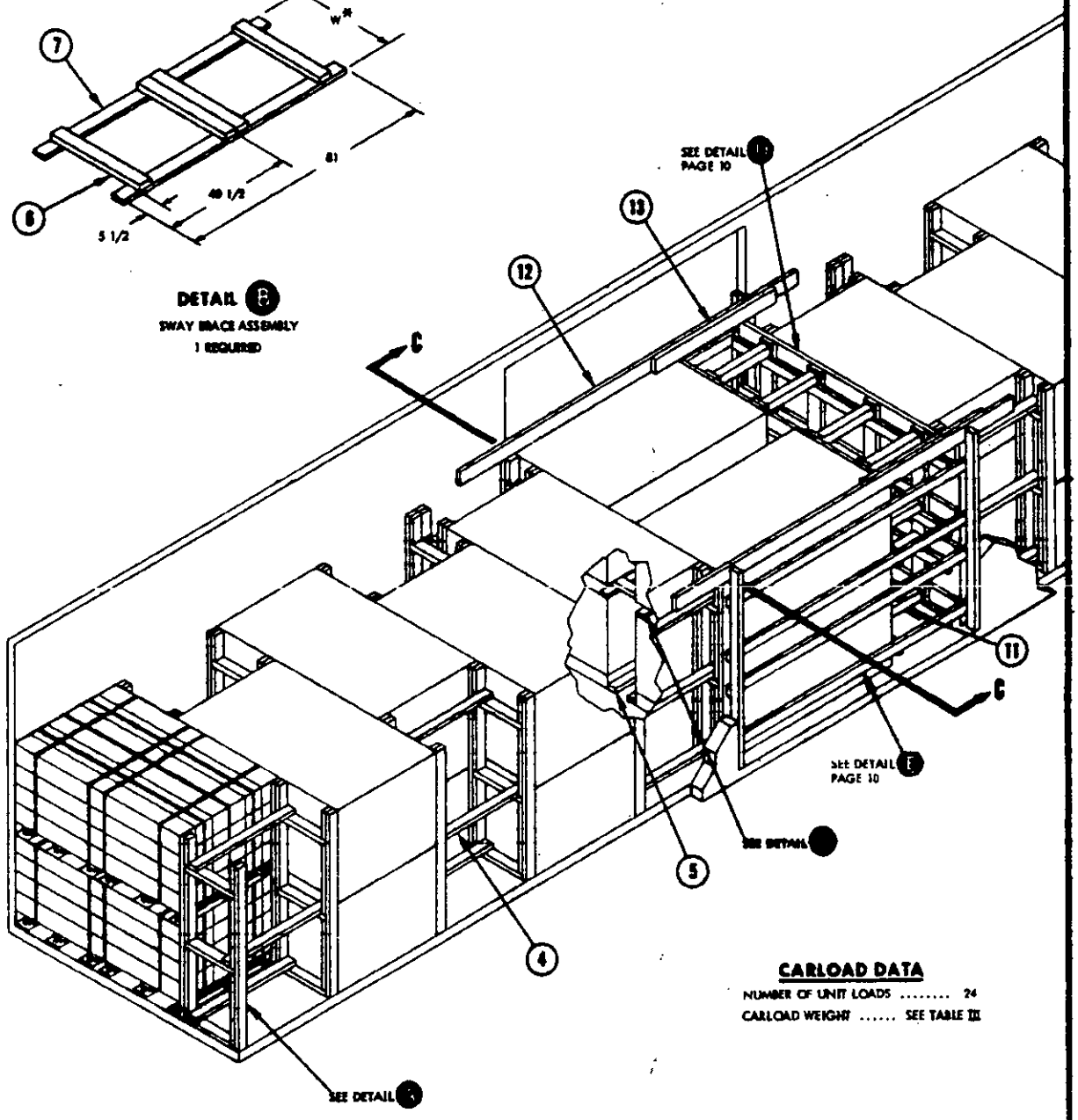
PIECE NO.	DESCRIPTION	SIZE	NO. PCS. REQD	NAIL TO	NUMBER NAILS	SIZE NAILS
15	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	8	14	3 PER JOINT	10d
14	VERTICAL DOORWAY MEMBER	2 x 3 x 84	4	DOOR POST	2 PER FOOT	20d
13	HOLD DOWN CLEAT	2 x 6 x 60	2	12	1 PER FOOT	10d
12	GATE HOLD DOWN	2 x 6 x 14 FT	2	CAR WALL	5 EA END	10d
11	STRUT	4 x 4 x WEDGE FIT**	16	9	2 PER JOINT	16d
10	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	9	3 PER JOINT	10d
9	CENTER GATE VERTICAL	2 x 6 x 76 3/4	8	SEE 8	-	-
8	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	9	3 PER JOINT	10d
7	STRINGER	1 x 4 x 81	2	6	3 PER JOINT	10d*
6	CROSS BRACE	2 x 4 x CUT TO FIT	4	SEE 7	-	-
5	SLEEPER	2 x 4 x 81	2	CAR FLOOR	2 PER FOOT	16d
4	SPREADER	2 x 4 x 49 1/2	36	2	2 PER JOINT	10d
3	SWAY BRACE CLEAT	2 x 4 x CUT TO FIT	144	1	3	10d
2	SWAY BRACE STRUT	2 x 4 x CUT TO FIT	72	SEE 1	-	-
1	SWAY BRACE VERTICAL	2 x 4 x 76 3/4	48	2	2 PER JOINT	16d

LIST OF MATERIALS AND NAILING DATA

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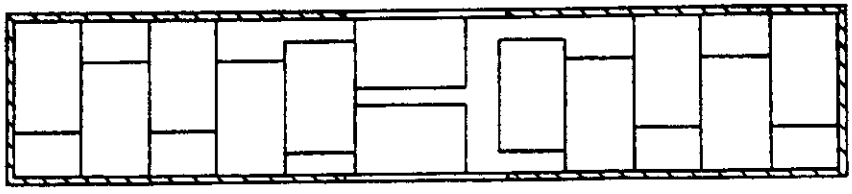
*W = DISTANCE BETWEEN POSTS OF PALLETS ACROSS THE CAR

DETAIL B
SWAY BRACE ASSEMBLY
1 REQUIRED



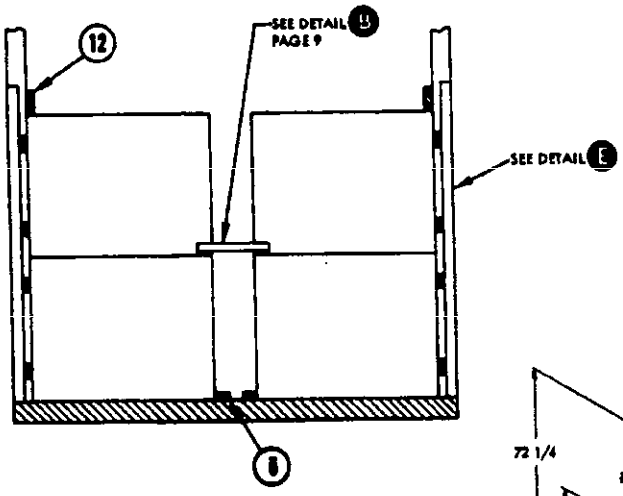
CARLOAD DATA

NUMBER OF UNIT LOADS 24
CARLOAD WEIGHT ..... SEE TABLE III

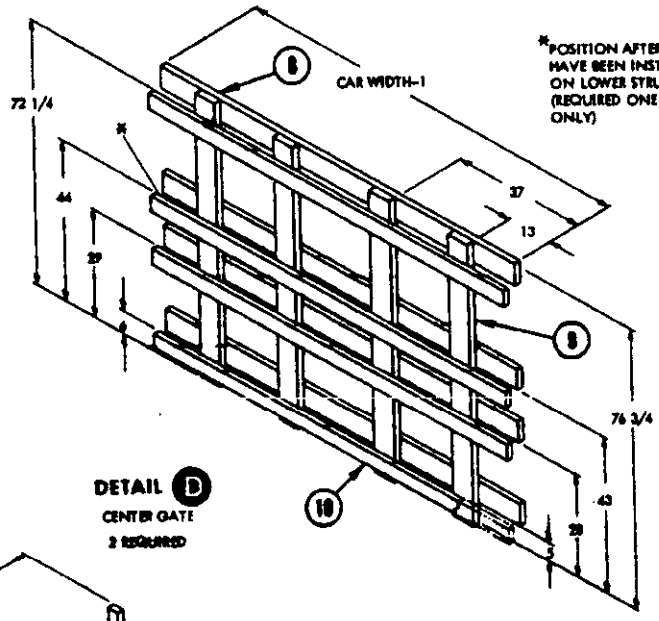


LOADING PLAN

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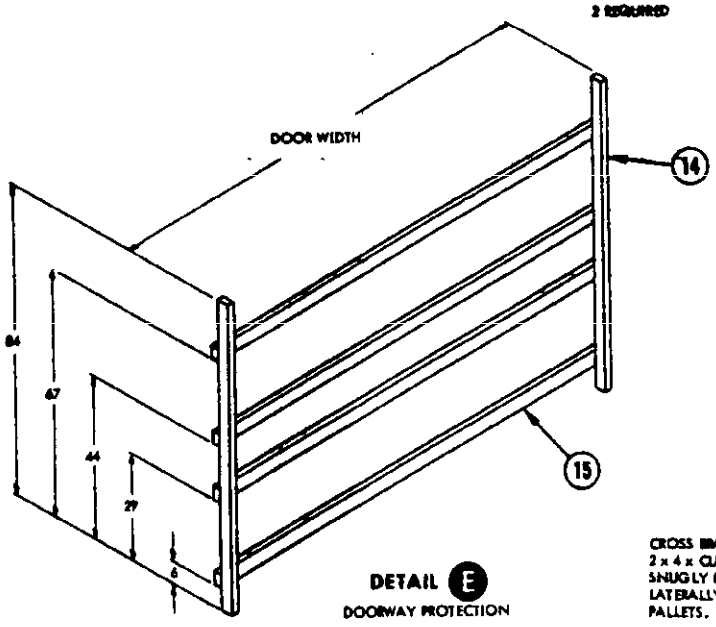


SECTION C-C



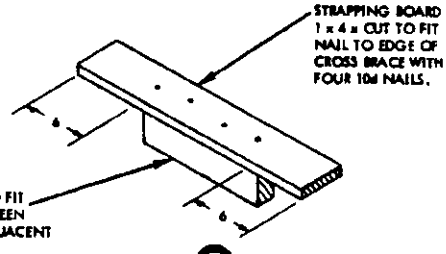
*POSITION AFTER STRUTS HAVE BEEN INSTALLED ON LOWER STRUT CLEAT (REQUIRED ONE GATE ONLY)

DETAIL D
CENTER GATE
2 REQUIRED



DETAIL E
DOORWAY PROTECTION
2 REQUIRED

WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) PAGE 46, FIG. 21 IS REQUIRED.



DETAIL F
TOP SWAY BRACE

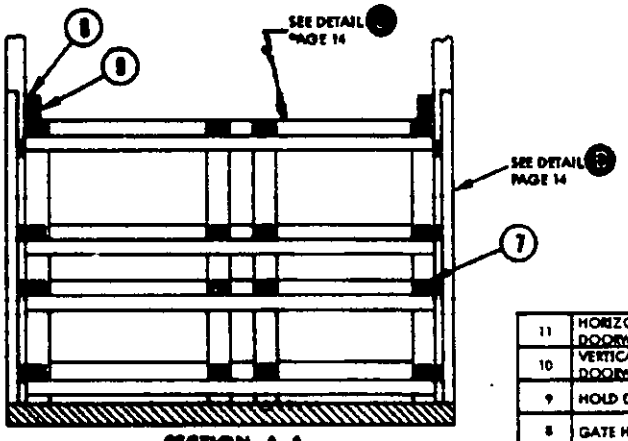
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50 FT 6 IN. BOXCAR, COMMERCIAL (ALTERNATE METHOD)

1. THE CARLOAD CONSISTS OF 28 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING. THE LOADS ARE HANDLED FROM THE END BY A FORK TRUCK WITH APPROVED FORK TINE EXTENSIONS OR A HAND PALLET TRUCK WHICH HAS THIS CAPABILITY.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH DETAILS PAGES 8, 9 AND 10 OF WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING THE LENGTH OF PIECE 8 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 8 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. THE STACKS THAT ARE IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND A TOP SWAY BRACE CONSISTING OF A STRAPPING BOARD AND CROSS BRACE (SEE DETAIL E) MUST BE INSTALLED BETWEEN THE ROWS OF THE DOORWAY STACKS. STRAP IS SECURED TO STRAPPING BOARD WITH TWO 1 1/4" STAPLES. DOORWAY PROTECTION, PIECES 10 AND 11 IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.
5. IF CAR HAS STEEL SIDEWALLS DELETE PIECE NUMBERS 8 & 9. TO HOLD GATE ASSEMBLY DOWN NAIL A DOUBLED 2 x 4 x 5 FT. HORIZONTAL TO THE GATE VERTICALS ON THE SIDE FACING THE LADING. POSITION HORIZONTALS WITH THE BOTTOM EDGE 39 INCHES FROM THE FLOOR (FIELD CHECK THIS DIMENSION) AND SO THAT THE HORIZONTAL IS TRAPPED UNDER THE UPPER UNIT LOAD. NAIL EACH JOINT WITH THREE 10d NAILS PER JOINT.



SECTION A-A

TABLE IX

ROCKET MOTOR	APPROXIMATE WEIGHT IN LBS	
	UNIT LOAD	CARLOAD
MK 16 & MODS	2741	77833
MK 71 MOD 0	2813	79849
APPROXIMATE WEIGHT OF DUNNAGE (2292 LBS) INCLUDED IN CARLOAD WEIGHT		

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
11	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	8	10	3 PER JOINT	10d
10	VERTICAL DOORWAY MEMBER	2 x 3 x 84	4	DOOR POST	2 PER FOOT	20d
9	HOLD DOWN CLEAT	2 x 6 x 24	4	8	5	10d
8	GATE HOLD DOWN	2 x 6 x 14 FT	2	CAR WALL	5 EA END	10d
7	STRUT	4 x 4 x WEDGE FIT *****	16	5	2 PER JOINT	16d
6	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	5	3 PER JOINT	10d
5	CENTER GATE VERTICAL	2 x 6 x 76 3/4	8	SEE 4	-	-
4	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	8	5	3 PER JOINT	10d
3	END PIECE	2 x 2 x CUT TO FIT **	28	2	2 PER JOINT	10d *
2	STRINGER	1 x 4 x 81	28	1	3 PER JOINT	10d *
1	CROSS BRACE	2 x 4 x CUT TO FIT	56	SEE 2	-	-

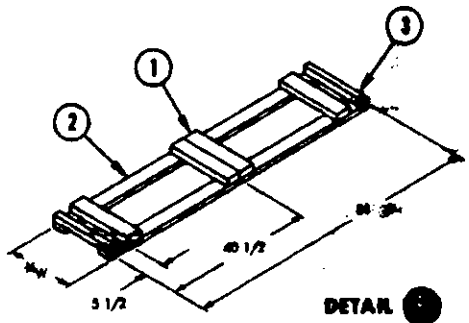
LIST OF MATERIALS AND NAILING DATA

* CLINCHED

** CAN BE RIPPED FROM 2 x 4

*** 2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4'S.

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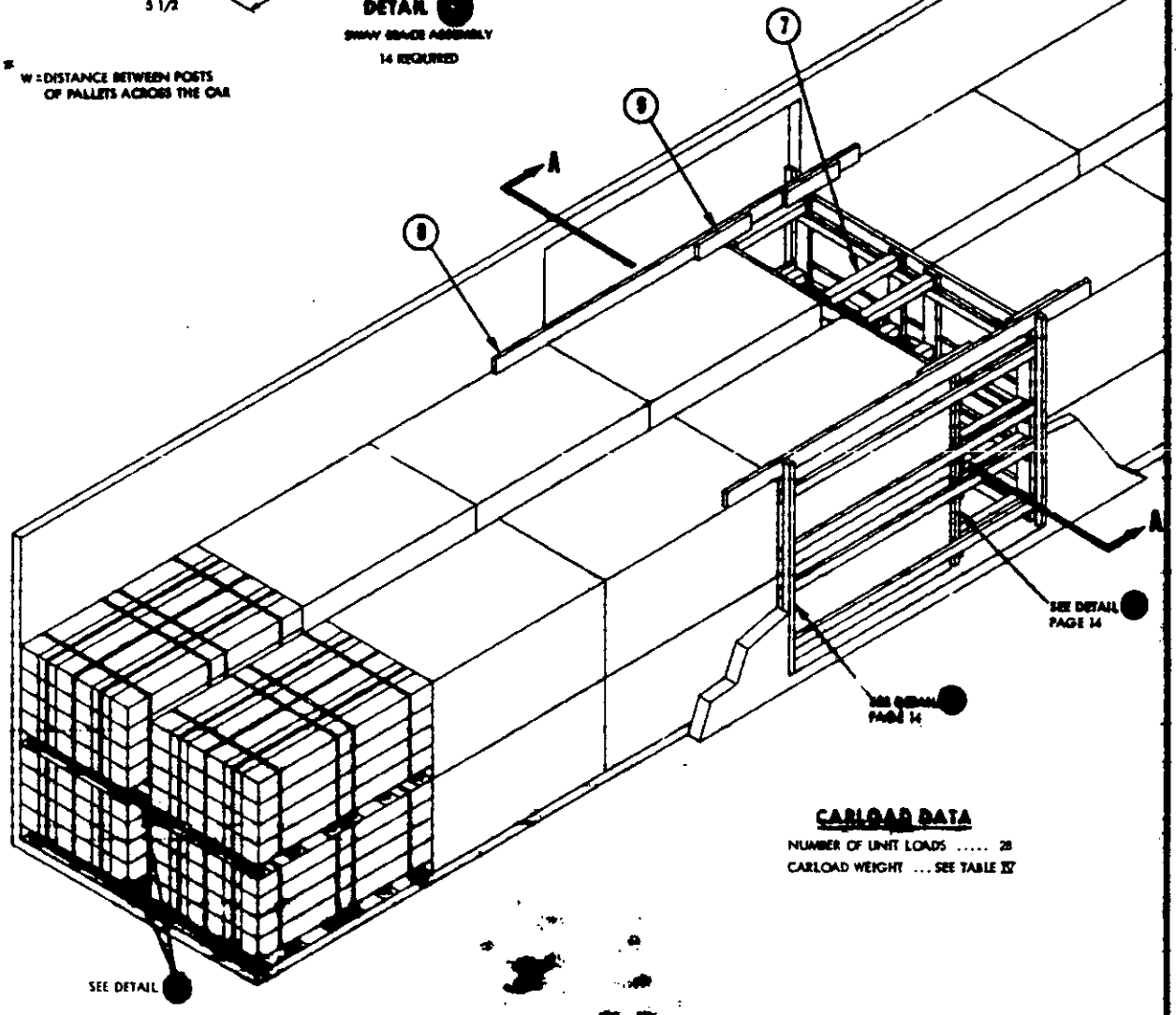


DETAIL

SHIM SPACE ASSEMBLY

14 REQUIRED

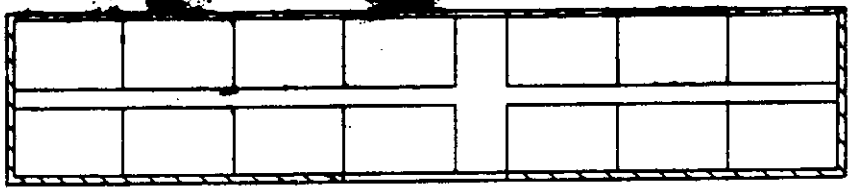
W = DISTANCE BETWEEN POSTS OF PALLETS ACROSS THE CAR



CARLOAD DATA

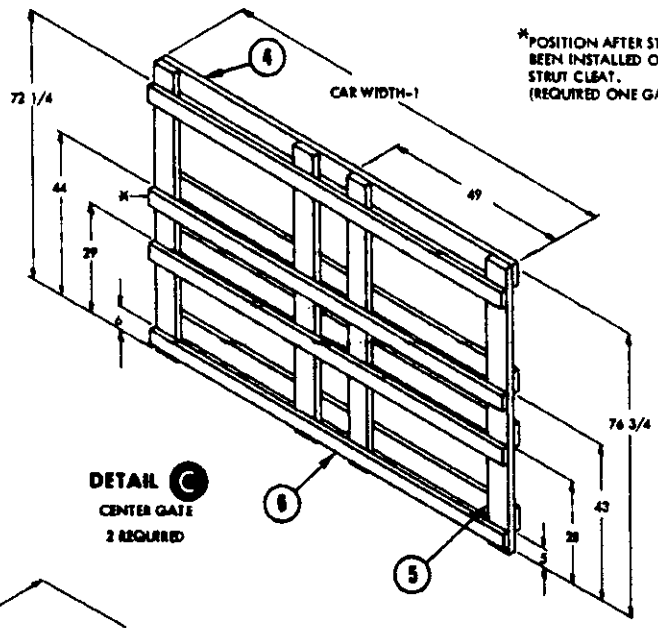
NUMBER OF UNIT LOADS 28

CARLOAD WEIGHT ... SEE TABLE IV



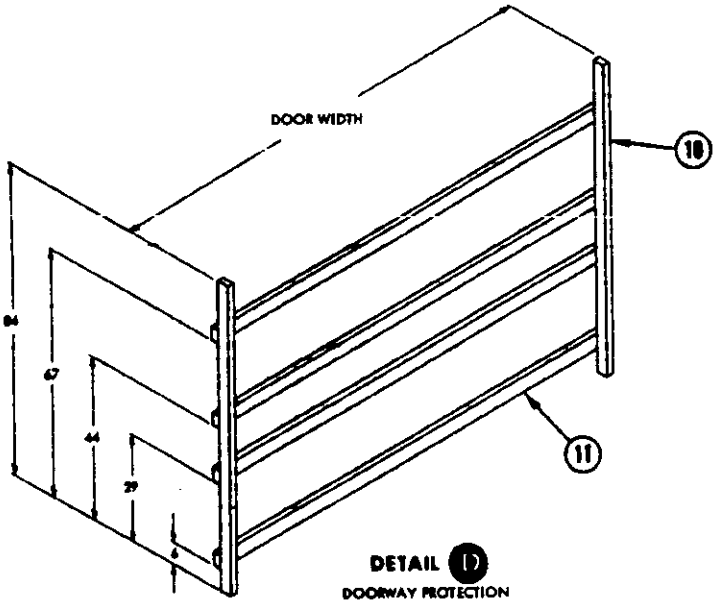
LOADING PLAN

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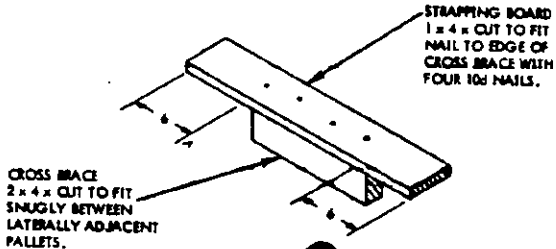
*POSITION AFTER STRUTS HAVE BEEN INSTALLED ON LOWER STRUT CLEAT. (REQUIRED ONE GATE ONLY)

DETAIL C
CENTER GATE
2 REQUIRED



DETAIL D
DOORWAY PROTECTION
2 REQUIRED

WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) PAGE 46, FIG. 21 IS REQUIRED.



DETAIL E
TOP SWAY BRACE

REVIEW ACTIVITY
NAVY - OS, AS

PREPARING ACTIVITY
NAVY - OS
(PROJECT NO. 8140 - N235)

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FOLD

[redacted] AND FEES PAID

[redacted]

[redacted]

Commanding Officer
Naval Ammunition Depot Earle
Colts Neck, New Jersey 07722

ATTN: NWHL/E. Connolly

FOLD

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-STD-1325-37 (NAVY)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N 0102-014-1801 6-25254

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